PROJECT ONE

Hybrid Time Series Forecasting

Improving retail demand forecasting using statistical, machine learning and hybrid models.

Forecasting is not about predicting the future with certainty — it is about making better business decisions with the information available today.

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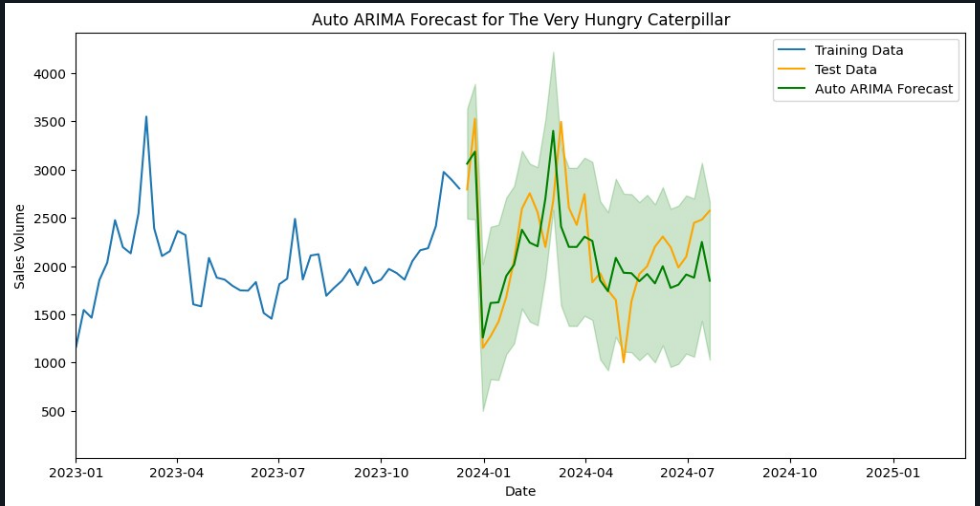
Data Analytics • Forecasting • Applied AI



Retail sales demand forecast

Classical and hybrid forecasting models compared across two sales datasets.

The business story in one page

A forecasting assignment reframed as a business-facing case study.

CHALLENGE

Forecasting demand is hard

Stock shortages, excess inventory and poor procurement decisions are often symptoms of unreliable demand forecasts.

APPROACH

Five model families tested

Auto ARIMA, XGBoost, LSTM, Sequential Hybrid and Parallel Hybrid models were compared using consistent evaluation metrics.

OUTCOME

Hybrid model led

The weighted Parallel Hybrid achieved the strongest weekly forecasting performance across both datasets.

BUSINESS VALUE

Better planning decisions

The approach can support procurement, reordering and inventory management decisions for retail SKU forecasting.

17.07%

Lowest weekly MAPE achieved on Dataset A using the Parallel Hybrid model.

Core finding

No single model wins everywhere. Hybrid forecasting offered the best balance of accuracy, robustness and practical value.

BUSINESS CONTEXT

Why forecasting matters

Accurate demand forecasting is one of the most influential drivers of retail performance.

INVENTORY

Maintain availability

Reduce excess stock while protecting product availability and customer service levels.

PROCUREMENT

Order with confidence

Purchase the right products at the right time based on expected demand patterns.

COST

Reduce waste

Lower storage costs and reduce capital tied up in unnecessary inventory.

SERVICE LEVEL

Avoid missed sales

Improve product availability and reduce lost revenue caused by stock-outs.

Project scope

This project investigates whether modern forecasting techniques can improve demand prediction using historical retail sales data. Two bestselling books were selected as representative time series because they exhibit different demand characteristics while sharing sufficient historical observations for meaningful model comparison.

Objective

Compare statistical, machine learning and hybrid forecasting approaches to determine which provides the strongest balance of accuracy and business applicability.

PROJECT WORKFLOW

From historical sales to forecast recommendations

A business-facing process, supported by technical forecasting and evaluation.

01	**Data preparation**	Merged Excel sheets, cleaned dates and ordered the time series.
02	**Exploratory analysis**	Identified trend, seasonality and sales behaviour from 2012 onwards.
03	**Model development**	Built statistical, machine learning, deep learning and hybrid models.
04	**Evaluation**	Compared MAE, RMSE and MAPE on unseen test data.
05	**Recommendation**	Translated results into inventory and forecasting guidance.

MODELS COMPARED

A practical test of model families, not a beauty contest

The aim was to understand where different approaches create value and where they fall short.

Auto ARIMA

STATISTICAL BASELINE

Interpretable, handles autocorrelation and provides a strong benchmark.

XGBoost

ML BENCHMARK

Flexible with lag features but does not inherently understand sequence.

LSTM

DEEP LEARNING

Designed for temporal patterns, but can smooth out seasonality on limited data.

Sequential Hybrid

TWO-STAGE HYBRID

Uses statistical structure first, then models residual patterns.

Parallel Hybrid

WEIGHTED ENSEMBLE

Combines forecasts to reduce reliance on any single method.

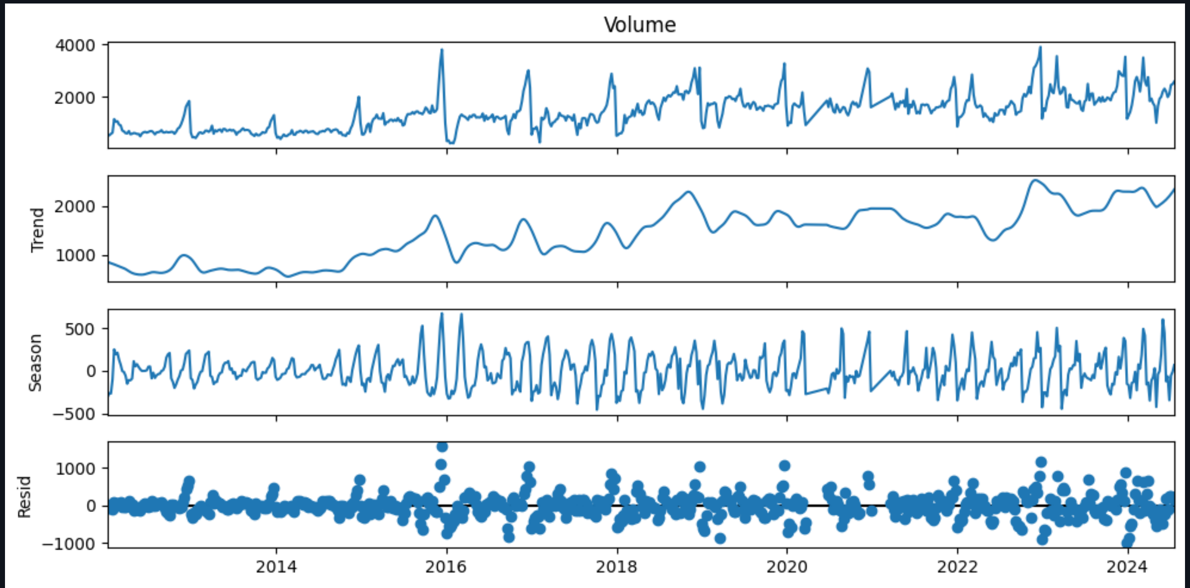
The hybrid approaches mattered because they tested whether combining strengths could produce a more reliable forecast than a single model family.

Trend and seasonality shaped the modelling approach

Decomposition highlighted recurring patterns, which made seasonal forecasting approaches relevant.

OBSERVED · TREND · SEASONAL · RESIDUAL

Dataset A decomposition



FINDING 01

Trend was visible

Long-term movement was present, so models needed to capture trend rather than treat demand as random.

FINDING 02

Seasonality mattered

Recurring demand patterns made seasonal forecasting approaches especially relevant.

FINDING 03

Noise remained

Residual variation showed why diagnostics and model comparison remained necessary.

RESULTS OVERVIEW

Hybrid forecasting delivered the strongest weekly performance

MAPE was used as the headline metric because it is easy to compare across datasets.

Model	Dataset A MAPE	Dataset B MAPE
Auto ARIMA	17.17%	25.79%
XGBoost	26.89%	34.93%
LSTM	23.85%	27.15%
Sequential Hybrid	17.13%	25.80%
Parallel Hybrid	17.07%	23.31%

Best weekly model

Parallel Hybrid

17.07%
Dataset A MAPE

23.31%
Dataset B MAPE

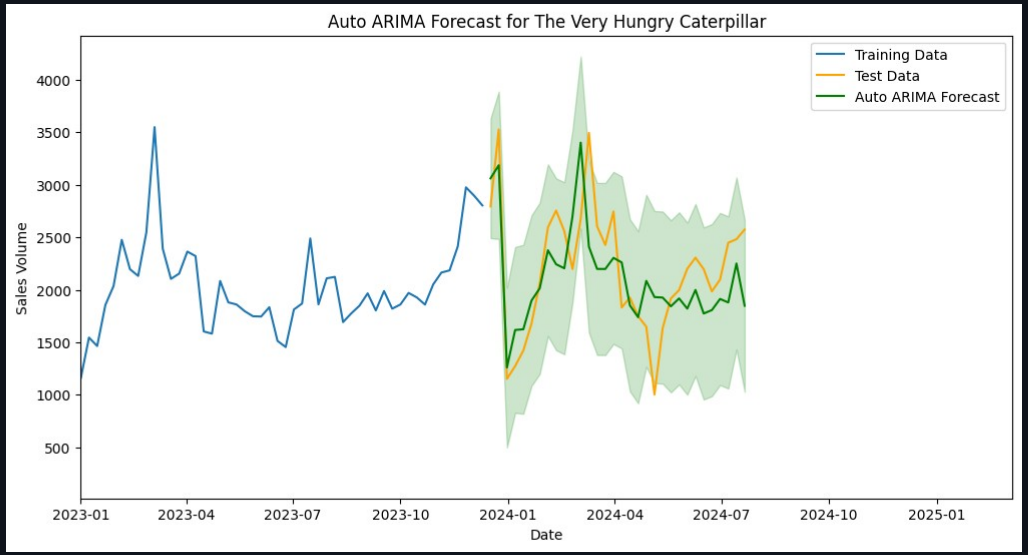
The result did not prove that complex models always win. It showed that combining model families can create more reliable forecasts than relying on a single technique.

The Very Hungry Caterpillar

Larger volumes and clearer seasonality made traditional and hybrid forecasting highly competitive.

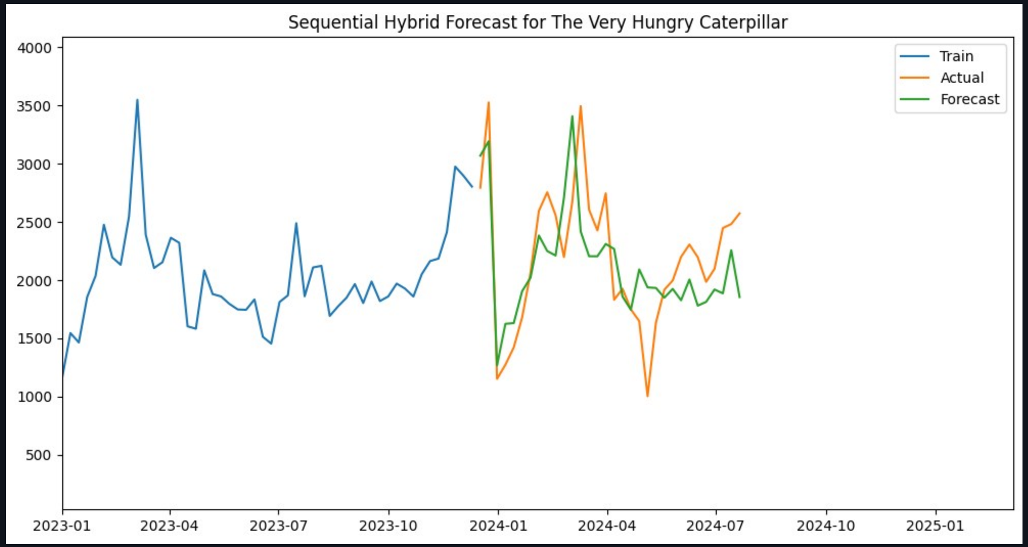
MAPE 17.17%

Auto ARIMA forecast



MAPE 17.13%

Sequential Hybrid forecast



Takeaway: statistical models captured the pattern well, while hybrid approaches slightly improved weekly accuracy and demonstrated the value of combining model strengths.

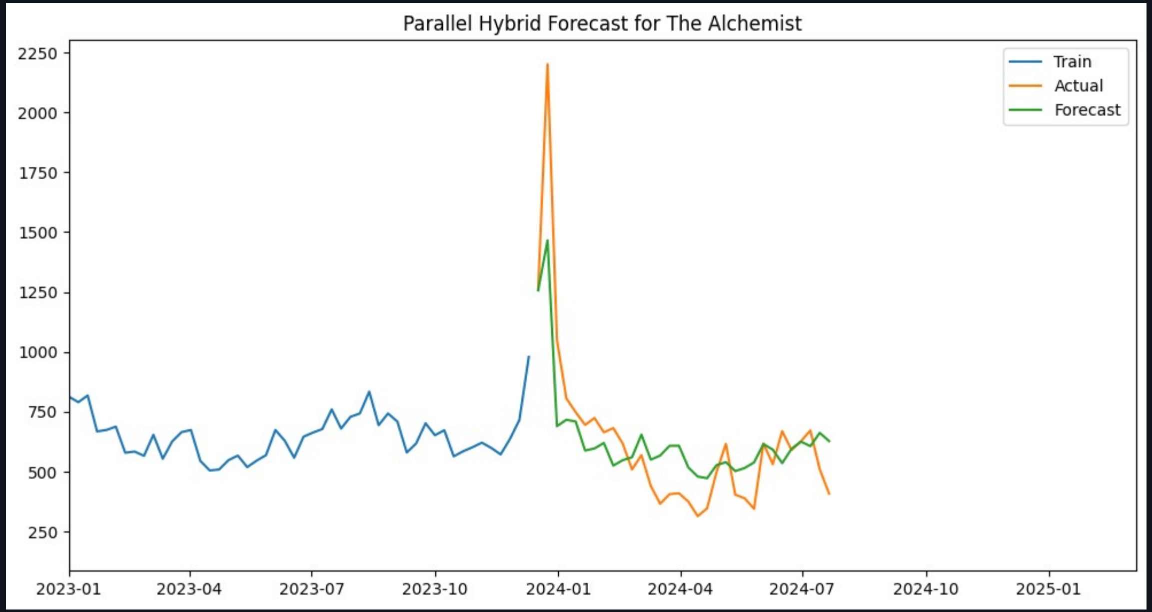
DATASET B

The Alchemist

Lower volumes and a more irregular demand profile made the series more challenging to forecast.

MAPE 23.31%

Parallel Hybrid forecast



OBSERVATION

Hybrid advantage was clearer

The Parallel Hybrid produced the strongest weekly MAPE for the Alchemist dataset, improving on Auto ARIMA and Sequential Hybrid performance.

INTERPRETATION

Different behaviour, different winner

The result reinforced that model selection should be based on the structure of each series rather than assuming one model will generalise everywhere.

Portfolio point

The strongest insight is not the model name — it is the evidence-led comparison.

How this work would translate into practice

The forecasting workflow can support retail decision-making beyond the original book sales dataset.

01

Use hybrid forecasting selectively

Prioritise hybrid methods for high-value SKUs where accuracy gains have the greatest commercial impact.

02

Monitor accuracy over time

Use rolling evaluation windows to detect when model performance starts to degrade.

03

Add business drivers

Incorporate promotions, holidays, price changes and marketing events as external variables.

04

Automate the workflow

Investigate AI-assisted forecasting pipelines for model selection, monitoring and reporting.

Recommended next implementation path

Prototype
Run on selected high-value SKUs

Monitor
Track MAPE / bias / drift

Improve
Add external drivers

Scale
Automate retraining & reporting

REFLECTION

What this project demonstrated

The strongest result came from comparing models carefully, not assuming complexity would win.

What surprised me

Classical forecasting remained highly competitive. Machine learning and deep learning models did not automatically outperform Auto ARIMA, especially where the time series had clear seasonal structure.

What worked

The strongest results came from combining techniques, using the strengths of statistical models and hybrid methods together.

NEXT FOCUS

Agents & LLMs for SKU Forecasting

I am now exploring how agents and LLMs could improve SKU-level demand forecasting by automating model monitoring, anomaly checks and business-facing summaries.

FUTURE CAPABILITY

AI-assisted forecasting

The natural next step is an agent-assisted workflow that helps select models, track forecast quality, flag unusual demand behaviour and translate outputs into business recommendations.

This portfolio case study is adapted from an academic forecasting project and rewritten to present the work in a business-focused format.